HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · CONSUMER DURABLES MANUFACTURING · PROCUREMENT

COPPER BUYING · 40+ SIGNALS · SIX-MONTH HORIZON

# Copper Price Intelligence

A GenAI-built cockpit put every copper price driver on one screen, forecast six months ahead, and turned a weekly email and gut feel into one rules-based buying call.

~~+1.2%~~ → **−1.9%** buy price vs LME monthly average

**−1.9 %** Copper buy price vs LME monthly average
▼ from +1.2% with reactive buying

**4.6 %** 3-month forecast error (MAPE)
▼ vs 7.9% for a no-change forecast

**2 hours** Weekly copper call prepared in
▼ from about 2 days of reports

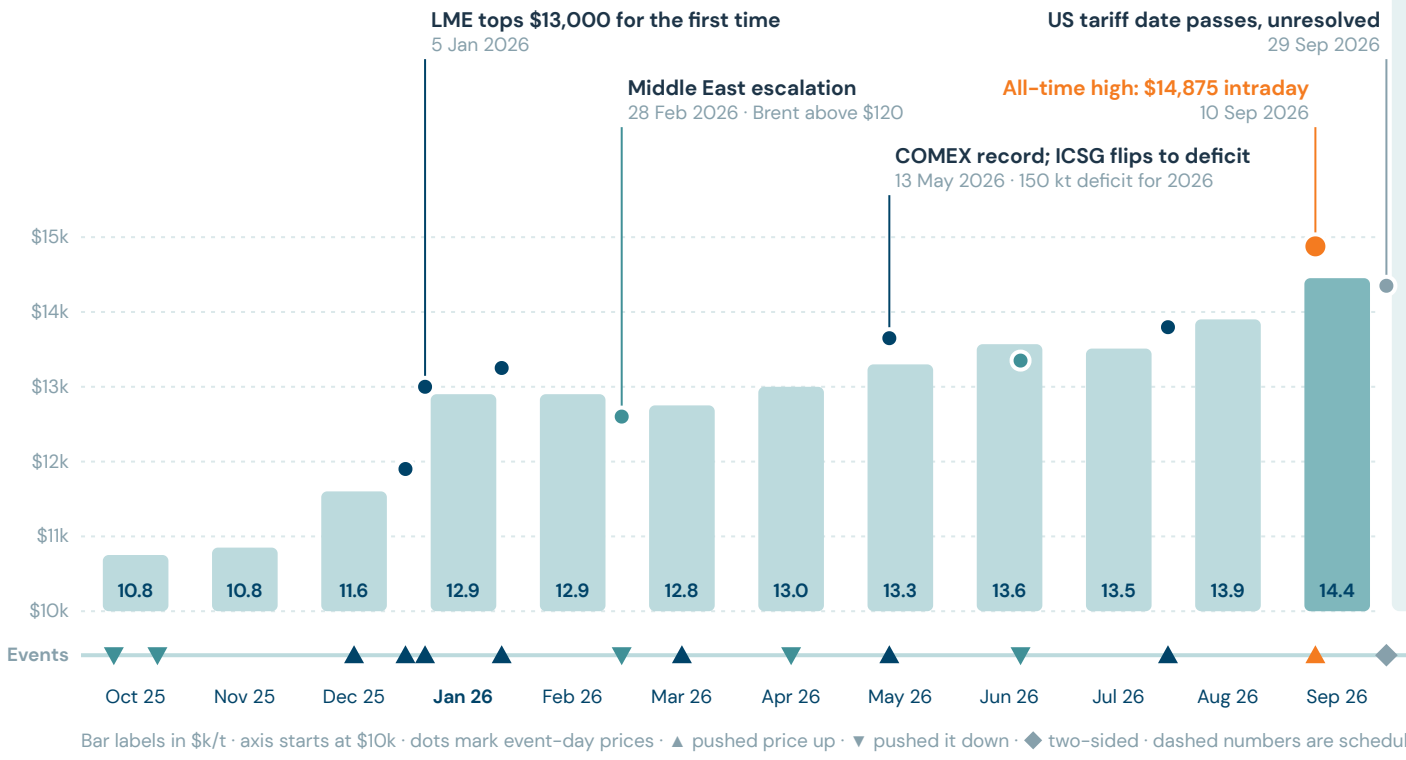
**40 + signals** Tracked every day
▲ 12 indices, 28 macro and micro factors

Rings: buy-price gap on a 0–5% scale (now outer, before inner); forecast error on a 0–13% scale with the no-change 7.9% inside; 2 hours on a 24-hour clock; one segment per signal (12 indices light, 28 factors dark). Figures indicative; validated metrics available on request.

## 01 Twelve months behind, six months ahead

Monthly LME copper price and the events that moved it · Oct 2025 – Mar 2027

PAST 12 MONTHS · MONTHLY LME AVERAGE, $/t · EVENTS THAT MOVED IT




NEXT 6 MONTHS · SCHEDULED EVENTS

1. **Mid Oct 2026** — LME Week London: 2027 premiums and outlooks
2. **Late Oct 2026** — FOMC meeting
3. **3 Nov 2026** — US midterm elections
4. **Dec 2026** — China economic work conference; 2027 TC/RC talks
5. **1 Jan 2027** — Proposed 15% US tariff on refined copper
6. **6 Feb 2027** — Chinese New Year: demand lull, then restock
7. **Mar 2027** — China NPC targets; India AC peak production

Bar labels in $k/t · axis starts at $10k · dots mark event-day prices · ▲ pushed price up · ▼ pushed it down · ◆ two-sided · dashed numbers are scheduled events, not predictions

Past events from public market reports (Cochilco, LME, CME, SHFE, ICSG, company releases and news coverage); monthly prices are indicative averages. Future pins are scheduled or expected events, not predictions.

## 02 Where copper's price is being made

Twelve factors on the signal map, ranked by model weight · plus the market pulse

| FACTOR · LAYER | WEIGHT IN MODEL | LATEST READING |
|---|---|---|
| **China demand & smelters** Demand | 20% ▲ | Physical premiums at 3-year highs; PMI 50.2; spot TC negative |
| **Chile mine output** Supply | 16% ▲ | Output -9% y/y in Mar 2026; Escondida -16%, Codelco -10% |
| **US tariffs & COMEX** Policy | 12% ◆ | Section 232: 15% on refined copper proposed from 2027; 29 Sep date passed unresolved |
| **Indonesia: Grasberg** Supply | 11% ▲ | Sep 2025 mudslide; 2026 guidance cut ~35%; full restart now 2028 |
| **Fed & the dollar** Macro | 8% ▼ | Hawkish June FOMC; DXY firmer at ~101 |
| **DRC: Kamoa-Kakula** Supply | 7% ▲ | Recovering from 2025 flooding; sulphuric-acid shortages limit output |
| **LME stocks & spreads** Market | 7% ▲ | Stocks ~250 kt, down from ~400 kt in April |
| **India demand & rupee** Demand / FX | 7% ▲ | AC and grid demand rising; ₹ at ~88.6 per $ |
| **Peru disruptions** Supply | 5% ▲ | Periodic road blockades on the southern mining corridor |
| **Strait of Hormuz** Geopolitics | 4% ◆ | Feb 2026 escalation sent Brent above $120 briefly |
| **Zambia expansion** Supply | 3% ▼ | New Copperbelt capacity ramping up |
| **Europe manufacturing** Demand | 3% ▼ | Factory activity soft; grid spend steady |

**LME COPPER 3M** ◆ Neutral
**$14,300** /t
+33% in 12 months; all-time high $14,875 intraday on 10 Sep 2026

**MCX COPPER** ◆ Neutral
**₹1,291** /kg
+36% in 12 months; the rupee reference, includes FX effect

**LME STOCKS** ▲ Up
**250** kt
Down from ~400 kt peak in April 2026; falling stocks support price

**COMEX STOCKS** ▼ Down
**770** k st
More than doubled in 12 months; metal pulled into the US on tariff risk

**CONCENTRATE TC** ▲ Up
**−$42** /t
Negative charges signal a very tight concentrate market

**SMM 1# PREMIUM** ▲ Up
**¥1,400** /t
Highest since Oct 2023; strong Chinese physical demand

**USD / INR** ◆ Neutral
**₹88.6**
A weaker rupee raises landed copper cost

**DOLLAR INDEX (DXY)** ▼ Down
**101.2**
Firmer since the hawkish June FOMC; weighs on metals

Tag = which way the reading pushes copper. Indicative snapshot for early October 2026, compiled from public market reports; the cockpit refreshes these from data feeds.

## 03 Four pain points, four fixes

How a copper buying decision gets made · before → after

BEFORE
### Record volatility
LME copper went from under $11,000 to nearly $14,900 a tonne in twelve months, with daily swings of 3–4%.

FORECAST A RANGE

AFTER
### Six-month range
"Prices look firm" became a six-month range with each driver's contribution shown.

BEFORE
### Drivers everywhere
A mudslide in Indonesia, a strike in Chile, a tariff memo, a PMI in Beijing. Nobody had the full picture.

SIGNAL MAP

AFTER
### 40+ factors, daily
One map refreshed daily, plus a calendar of scheduled events six months ahead.

BEFORE
### Reactive buying
Tubes and cathode bought to the production calendar, often above the monthly LME average in peak AC season.

HEDGE RULES

AFTER
### Laddered buys
Buy timing and hedge ratio from clear rules, agreed and logged each week.

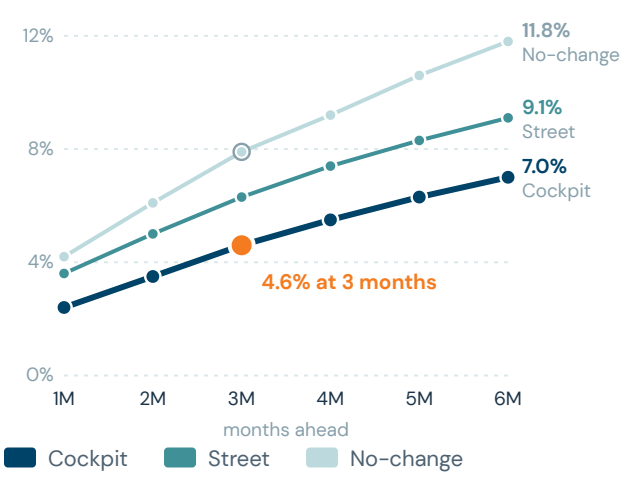
BEFORE
### Cost impact unclear
Sales and finance asked what a $500 move meant per AC and per quarter. The answer took days.

AC COPPER MODEL

AFTER
### Per-AC impact, live
Per-AC and annual impact in the cockpit; sales sees it before price revisions.

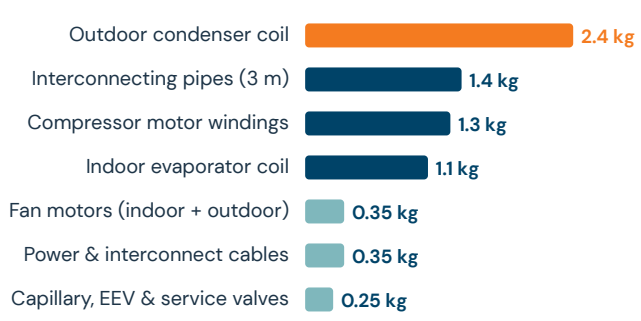
## 04 Forecast error

Mean absolute % error by horizon · back-test Jan 2024 – Sep 2026




## 05 Copper inside an AC

Typical copper by part, 1.5 TR inverter split AC




**7.3** kg — copper in one 1.5 TR inverter split AC; the condenser alone is a third of it

Typical industry values; vary by design and efficiency rating.

## 06 How HMC ran it

| | |
|---|---|
| 2 weeks | **Discover** Mapped copper exposure by product line; two years of buys reviewed against LME and MCX. |
| 2 weeks | **Design** Three-layer factor model; weights agreed with the CPO and treasury; hedge rules drafted. |
| 4 weeks | **Build** Claude turns daily news into factor signals; a transparent model does the maths. |
| 1 quarter | **Adopt** A 30-minute Monday copper call replaced scattered emails. |
| ∞ ongoing | **Scale** Aluminium added; steel and plastics next; accuracy reviewed quarterly. |

## What transfers to any commodity buyer

**01 — Put the drivers on a map**
When everyone can see the mine, the tariff and the PMI on one screen, the debate moves from opinions to weights.

**02 — Forecast a range, decide a rule**
No model calls the exact price. A range plus a hedge rule beats a single number and a gut call.

**03 — Translate $/t into ₹ per product**
Showing copper per AC turned a market story into a pricing and margin conversation everyone could join.

Built with **Claude** · **Python** · **D3.js** · **Three.js** · exchange and macro feeds | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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